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[Notes]
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Currently I'm studying representation theory and preparing my Master thesis with Prof. Dr. Raphaël Beuzart-Plessis and Prof. Dr. Vytautas Paškūnas. During my bachelor, my advisor was Prof. Dr. Shanwen Wang.

Current (Research) Interests

- **Representation Theory** and **Number Theory**, in particular Automorphic Forms and Trace Formula. Bump's book led me into this field.

Education

- **ALGANT** Master, full scholarship 2024.9 – Present
  Università di Padova (M1) and Universität Duisburg-Essen (M2, currently)
- Visiting Research Student, BIMCR, 2023.8 – 2024.6
- Enhanced program for Graduate Study (visiting research student), BIMCR, 2023.2 - 2023.7
- Undergraduate, Mathematics. 2020.9 – 2024.6

Teaching Assistant

- Modular Forms (Graduate course), 2023.9 – 2024.12
- Mathematical Analysis III (2rd year course), 2023.9 – 2024.12
- Calculus (1st year course), 2023.9 – 2024.12
- Real Analysis (2nd year course). 2023.2 - 2023.6

Master Thesis

In this note, we study a homological approach to the representation theory of p-adic groups, based on the work by Peter Schneider- Ulrich Stuhler and Ralf Meyer. To be precise, Meyer proved that the canonical embedding of the Hecke algebra into the Harish–Chandra Schwartz algebra is "isocohomological". Equivalently, restriction induces a fully faithful functor from the derived category of smooth tempered representations to the derived category of all smooth representations.

We simplify the proof of known comparison theorems for these two derived categories, due to Meyer. Our method is based on the Bruhat–Tits building of G and on analytic properties of the Schneider–Stuhler resolutions.

Preprints

[Parameters and theta lifts], with Zhe Li, Shanwen Wang

In this note, we make explicit the correspondence between Harish-Chandra parameters and Langlands-Vogan parameters for symplectic groups and orthogonal groups of equal rank over reals. As an application, we reformulate Moeglin's results and Paul's work on the Howe correspondence for symplectic-orthogonal dual pairs using Langlands-Vogan parameters.

In another paper, we will try to use the local trace formula approach to complete the proof of the local Gan-Gross-Prasad conjecture, symplectic-metaplectic case.

Bachelor Thesis

[Local Gan-Gross-Prasad Conjecture: Tempered Hermitian Case], 2023.4 - 2024.4

In this paper, we study, following work of Beuzart-Plessis in [BP20], a geometric expansion for certain multiplicities m(\pi) appearing in the local Gan-Gross-Prasad Conjecture, and Beuzart-Plessis's proof on a weak form of this conjecture (on the multiplicity one property in tempered L-packets). In his proof, the crucial step is to study a sort of local relative trace formula which is related to the Gan-Gross-Prasad Conjecture for unitary groups over a local field F of characteristic zero.

Selected Attended Mini Courses/Seminars

- Around the Langlands Program, 1st week and 5th week. 2026.1-2
- Aarhus Automorphic Forms Conference: Summer School 2025.8
- Representation Theory & Noncommutative Geometry 2025.2 – 2025.3

  I join in the [Workshop 2: Tempered representations and K-theory], p-adic week and Theta correspondence week in person. Details and Personal Notes can be found [Here].
- From Modular Forms to Shimura Varieties: an Introduction 2024.1
- Arizona Preliminary Winter School, Arizona (Online), 2023.9 – 2023.12

  Focus on the CM elliptic curve and class field theory of imaginary quadratic fields.
- Core topics in modern number theory, YMSC, 2022.12 - 2023.6

Selected Talks/Seminars

- Master number theory seminar: Iwasawa theory, 2025.10 – 2026.1
- Master number theory seminar: Prismatic Cohomology, 2026.4 – 2026.7
- Iwasawa theory and more on number theory study group, 2025.1 – 2025.6
- Reps theory on the Archimedean place with emphasis on GL(2), 2023.8 – 2024.4
- Algebraic Geometry and Elliptic Curves study group, 2022.9 – 2023.10
- Category Theory and Homological Algebra study group, 2023 Spring